HACK IN HILLS '26 | PROBLEM 03: COMMODITY DERIVATIVES INTELLIGENCE | TEAM LNX

GOLD INTELLIGENCE

Video Demo Scripts, Website Navigation Blueprint & Defense Playbook

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> 💡 **THE 30-SECOND MENTAL MODEL (How to Understand & Explain This Project Easily)**
>
> If anyone asks you what this project does in plain English, use this fruit market analogy:
>
> *"Imagine you visit a market where one vendor sells **10 kg boxes of 95% pure gold** quoted per 10 grams, while another vendor sells **1 gram packets of 99% pure gold** quoted per gram. If you just look at the screen prices, you are comparing apples to oranges!*
>
> ***Gold Intelligence** does 3 simple things:*
> ***1. Purity Normalization:** Converts every single contract into **₹ per 1.000 gram of 100% pure gold**.*
> ***2. Anomaly Detection:** Watches the relative spread and flags when one contract becomes abnormally expensive or cheap compared to historical norms.*
> ***3. The Reality Check:** Rejects fake 'paper profits' that would get wiped out by Indian exchange taxes (CTT, GST, Stamp Duty) or illiquid market depth."*

1. Pre-Recording Setup & Technical Checklist

Step 1: Start the Local Application

Open two terminal tabs in the project root directory:

- **Terminal 1 (Backend):** `cd backend && uvicorn app.main:app --port 8000 --reload`
- **Terminal 2 (Frontend):** `cd frontend && npm run dev`
- Open browser at: `http://localhost:5173`

Step 2: Recording Environment Setup

- **Resolution:** Set browser to 100% zoom at 1920x1080 (press `F11` for full-screen mode).
- **Audio:** Use a clear headset or USB mic. Speak with an analytical, calm pace. Pause for 1 second after stating key numbers.
- **Mouse Cursors:** Move your mouse deliberately 1 second *before* speaking about an element. Avoid rapid shaking or circular gestures.
- **Initial State:** Start recorded video on the **Overview** tab with the chart loaded on `GOLDM vs GOLDPETAL`.

2. Master Demo Video Script (4 to 5 Minutes)

Target Audience: Hackathon Judges, Technical Evaluators, and Comprehensive Project Reviews.

SCENE 1: The Hook & Problem Statement 0:00 – 0:45

> 🖥️ **WHERE TO NAVIGATE:**
> • Start on **Overview** page (`tab=overview`).
> • Point cursor at the top research banner: "*Is one MCX gold contract behaving unusually relative to another?*"
> • Slowly hover cursor across the 4 top metric cards: `Contracts Covered (4)` , `Historical Observations (722)` , and `Largest Standardized Deviation (9.18σ)` .

🗣️ **WHAT YOU SAY (SPOKEN WORD):**

"Hello judges! When people trade gold on MCX India, retail traders usually make a dangerous assumption: they try to guess whether gold will go UP or DOWN.

But institutional quantitative desks look at gold completely differently. On MCX, gold is listed across four distinct contracts: **GOLDM**, **GOLDTEN**, **GOLDGUINEA**, and **GOLDPETAL**.

Here's the catch: they all have different lot sizes, different quoting units, and different purity levels. You cannot compare their raw prices on screen!

My team, **Team LNX**, built **Gold Intelligence** — an institutional-grade quantitative relative-value and arbitrage platform that normalizes multi-contract gold derivatives, scans for statistical mispricings, and proves whether they survive real-world Indian transaction costs."

SCENE 2: Contract Explorer & Purity Normalization Engine 0:45 – 1:30

> 🖥️ **WHERE TO NAVIGATE:**
> • In the left sidebar, click **Contract Explorer** (`tab=contracts`).
> • Point cursor to the **GOLDM** card (100g lot, quoted per 10g, **995 purity**).
> • Point cursor to the **GOLDPETAL** card (1g lot, quoted per 1g, **999 purity**).
> • Hover over the formula banner at the top: `P_fine = (P_raw / QuotationUnit) × (1000 / Purity)` .

🗣️ **WHAT YOU SAY (SPOKEN WORD):**

"Let's look at the contract specifications to understand why naive comparisons fail.

Notice here: **GOLDM** is quoted in ₹ per 10 grams at 995 fineness. Meanwhile, **GOLDPETAL** is quoted in ₹ per 1 gram at 999 fineness.

If GOLDM is trading at ₹75,000 and GOLDPETAL is trading at ₹7,550, is one actually cheaper than the other?

You can't tell until you normalize them. Our platform runs a deterministic mathematical conversion that scales every single tick into an identical benchmark: **₹ per 1.000 gram of 1000-fine pure gold**.

Once every contract is on the exact same footing, we can calculate the true relative spread in basis points."

SCENE 3: Live Relative-Spread Chart & Anomaly Detection 1:30 – 2:30

> 🖥️ **WHERE TO NAVIGATE:**
> • Click back to the **Overview** tab in the sidebar (`tab=overview`).
> • Scroll down to the visual centerpiece: the **Interactive Relative Spread Chart**.
> • Slowly hover cursor along the blue spread line to showcase the interactive tooltip: `Date` , `Spread (bps)` , `Baseline (μ)` , and `Z-score` .
> • Toggle the contract pair dropdown from `GOLDM vs GOLDPETAL` to `GOLDTEN vs GOLDGUINEA` and back.
> • Point out the **red dots** (statistical dislocations) and the **shaded blue band** (±2σ confidence envelope).

🗣️ **WHAT YOU SAY (SPOKEN WORD):**

"Now let's look at the heart of our platform — the Relative Spread Engine.

This blue line is the synchronous basis spread between GOLDM and GOLDPETAL, expressed in basis points.

The dashed line is our trailing 20-day rolling mean, and the light blue envelope represents dynamic **±2 standard deviation Bollinger-style confidence bands**.

Because commodity volatility changes across market regimes, our confidence bands expand dynamically during macro volatility and tighten during calm periods.

Look right here [hover over the large spike]: we detected a massive dislocation where GOLDM diverged by **9.18 standard deviations** away from its historical mean!

In pure statistics, that is an extreme event. But in quantitative finance, we must ask: *can an analyst actually trade this, or is it a trap?* That brings us to our core innovation."

SCENE 4: The Anomaly Scanner & Triple-Filter Reality Gate 2:30 – 3:30

> 🖥️ **WHERE TO NAVIGATE:**
> • In the sidebar, click **Anomaly Scanner** (`tab=anomalies`).
> • Point cursor to the filter summary cards: `Liquidity Gate` , `Expiry Gate` , `Cost Hurdle` .
> • Click on any anomaly finding row (or the 'Inspect' button) to pop open the **Finding Detail Modal** showing the tax breakdown.

🗣️ **WHAT YOU SAY (SPOKEN WORD):**

"Most algorithmic projects stop at finding a 2-sigma deviation. But in real-world Indian commodity trading, **over 75% of statistical anomalies are unexecutable traps**.

Why? Because they happen during illiquid hours with wide bid-ask spreads, or near contract expiry during physical tender periods, or the spread is simply too small to cover exchange taxes.

To solve this, we engineered the **Triple-Filter Execution Gate**:

1. Gate 1 (Liquidity): Requires both contracts to have daily volume above 100 lots and open interest above 500. This wipes out ghost prints.
2. Gate 2 (Expiry Tender Safety): Rejects trades if expiry is within 5 trading days, completely protecting traders from MCX compulsory physical delivery margins.
3. Gate 3 (Statutory Cost Hurdle): We model the full Indian tax schedule — Commodity Transaction Tax (0.01%), Stamp Duty (0.002%), MCX Turnover Fees, 18% GST, and 3.5 basis points of execution slippage per leg.

A signal is ONLY promoted to a tradable opportunity if the expected reversion spread strictly exceeds our **14.0 basis point round-trip cost hurdle**."

SCENE 5: Walk-Forward Backtesting Lab & Beta = 0.002 Proof 3:30 – 4:25

> 🖥️ **WHERE TO NAVIGATE:**
> • In the sidebar, click **Backtesting Lab** (`tab=backtest`).
> • Click the blue **"Run Backtest"** button (play icon) to show the simulation executing live.
> • Point directly at the scorecard badges on the right:
> • **Beta to Underlying Gold: 0.002** (Green badge).
> • **Correlation to Gold: 0.051**.
> • **Total Statutory Frictions Deducted** (shows ₹16,392 deducted).
> • Hover over the **Equity Curve Chart**: show Gross Model (blue line), Net Post-Friction Trajectory (green line), vs the volatile Directional Gold Benchmark (grey line).

🗣️ **WHAT YOU SAY (SPOKEN WORD):**

"Now let's verify if this strategy actually works historically. We built a walk-forward chronological backtesting engine.

Crucially, our backtest has **zero look-ahead bias**. Signals detected on day t close are executed strictly on day t+1 open.

Look at these empirical results on screen:

Our strategy's **Beta to underlying gold is 0.002**, and the **Correlation to gold is just 0.051**.

What does that mean? It means our alpha is **100% market neutral**. Whether gold crashes 20% or rallies 30%, our portfolio is immune to directional swings! Our return comes purely from relative-value convergence.

Furthermore, notice that our equity curve transparently accounts for **every single rupee of CTT, GST, Stamp duty, and slippage**. We don't hide costs; we show exactly what survives in the real world."

SCENE 6: Architecture, 25/25 Pytest Tests & Closing **4:25 – 5:00**

> 🖥️ **WHERE TO NAVIGATE:**
> • Click briefly to **Methodology** or **Data Center**.
> • Return to **Overview**. Point to the top telemetry indicator: `Status: Online · 722 Observations`.

🗣️ **WHAT YOU SAY (SPOKEN WORD):**

"Under the hood, Gold Intelligence is built with a high-performance **FastAPI backend**, an institutional **SQLite repository** with 722 synchronous daily observations, and a clean **React/Tailwind terminal**.

Our codebase includes **25 out of 25 passing automated unit tests** verifying our normalization formulas, cost hurdles, and portfolio execution.

This is not a static UI mockup. It is a fully functioning quantitative commodity intelligence workstation built by Team LNX for Hack in Hills '26.

Thank you so much for your time, and we look forward to your questions!"

3. The 60-Second Viral Video Guide (LinkedIn, X, Reels)

High-energy, zero fluff, fast cuts, optimized for mobile viewing and tech engagement.

Timestamp	Where to Click & Navigate	🗣️ Voiceover Script
0:00 – 0:10	Start on **Overview**. Move mouse deliberately across contract names: `GOLDM`, `GOLDTEN`, `GOLDGUINEA`, `GOLDPETAL`.	*"Did you know that on the MCX exchange, gold contracts don't actually cost the same? Here is why retail traders get trapped."*
0:10 – 0:22	Click **Contract Explorer**. Point to GOLDM 100g (995) vs GOLDPETAL 1g (999). Hover over normalization formula.	*"GOLDM is quoted per 10g at 995 purity. GOLDPETAL is quoted per 1g at 999 purity. Comparing raw prices is an apples-to-oranges illusion. We built **Gold Intelligence** to convert every contract into **₹ per pure fine gram**."*
0:22 – 0:38	Click **Overview**. Hover over the spike on the **Relative Spread Chart** with the red anomaly dots.	*"Once normalized, we track basis point spreads in real-time. Look at this spike: a massive **9.18 standard deviation dislocation** between contracts! When that happens, one contract is artificially cheap while the other is expensive."*
0:38 – 0:50	Click **Backtesting Lab**. Click **Run Backtest**. Point cursor directly at **Beta = 0.002**.	*"We run it through our Triple-Filter Gate, deducting Indian CTT, Stamp Duty, GST, and slippage. In backtesting, our strategy achieved a **Beta to Gold of 0.002** — pure market-neutral alpha with zero directional risk!"*
0:50 – 1:00	Zoom out slightly or show the entire sleek terminal UI.	*"100% built with FastAPI, SQLite, React, and 25 passing tests. Built for Hack in Hills '26 by Team LNX. Check out the project links below!"*

4. Copy-and-Paste Social Media Captions

A. Professional Post for LinkedIn

Did you know that on MCX India, gold futures don't actually trade at the same price? 🪙📈

Most retail traders focus on guessing whether gold will go UP or DOWN. But institutional quantitative desks look at something else entirely: Relative Value.

MCX lists 4 gold contracts: GOLDM, GOLDTEN, GOLDGUINEA, and GOLDPETAL.
Here's the trap:
❌ Different lot sizes (100g down to 1g)
❌ Different quoting bases (₹/10g vs ₹/1g vs ₹/8g)
❌ Different purity grades (995 vs 999 fineness)

Comparing raw screen prices is an apples-to-oranges illusion!

For Hack in Hills '26, our team (Team LNX) built **Gold Intelligence** — an institutional quantitative relative-value and arbitrage platform:

🔍 What we built:
1️⃣ Purity Normalization Engine: Deterministically converts every contract quote into ₹ / 1.000g of 1000-fineness gold.
2️⃣ Statistical Anomaly Scanner: Tracks basis spread dynamics with dynamic ±2σ volatility envelopes (detecting dislocations up to 9.18σ!).
3️⃣ The "Triple-Filter Gate": Over 75% of statistical anomalies are unexecutable traps. We enforce strict liquidity depth, near-expiry tender safety, and a statutory cost hurdle (deducting CTT, Stamp Duty, MCX turnover fees, GST, and slippage).
4️⃣ Walk-Forward Backtester: Zero look-ahead bias with t+1 next-bar execution.

📊 The Key Quant Result:
Our strategy generated a Market Beta to Gold of just $\beta = 0.002$ (Correlation $\rho = 0.051$) — proving true market neutrality. Returns are uncorrelated with directional commodity bull or bear cycles.

Tech Stack: Python, FastAPI, NumPy/Pandas, SQLite, React, Tailwind CSS, Recharts. Fully covered with 25/25 passing Pytest test suites.

Built by Team LNX: Rudra Sharma, Ayush Argulwar, Rohit Thoke.

Check out the full demo video and GitHub repo below! 👇

#Fintech #QuantitativeFinance #CommodityTrading #AlgorithmicTrading #Python #FastAPI #React #Hackathon #MCX #HackInHills26

B. Punchy Post for X (Twitter)

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Why do retail traders lose money trading MCX Gold spreads? 🪙💣

They compare raw screen prices between GOLDM & GOLDPETAL.
Big mistake: one is quoted per 10g (995 purity), the other per 1g (999 purity).

We built Gold Intelligence to solve this for Hack in Hills '26.

Here's how it works 👇

1/ 📐 Purity Normalizer: Maps all 4 MCX gold symbols to a single canonical standard: ₹ per 1.000g of pure gold.

2/ ⚡ Anomaly Scanner: Computes rolling basis spreads with dynamic ±2σ envelopes. We spotted historical
dislocations as wide as 9.18σ!

3/ 🛡️ The Reality Gate: Paper arbitrage is useless if taxes eat it. We filter out trades that don't survive
Indian CTT, Stamp Duty, MCX fees, GST & slippage.

4/ 📊 Empirical Proof: Next-day (t+1) walk-forward backtest achieved β = 0.002 to gold. Pure market-neutral
alpha independent of gold market direction.

Tech: FastAPI + SQLite + React + 25/25 Pytest passing.

Built by Team LNX (@HackInHills). Full demo in the video! 🚀
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5. Tough Judge Q&A Cheat Sheet (Defense Playbook)

Q1: "Why relative value instead of predicting directional gold price?"

Your Answer: *"Directional speculation is exposed to macro events, interest rate decisions, and geopolitical shocks. Relative-value is market-neutral. Because contracts represent the same physical commodity, their normalized prices must converge by expiry. By trading the basis spread, our beta to gold is 0.002 — we don't care whether gold crashes or rallies."*

Q2: "Isn't this just a Bollinger Band on a price spread?"

Your Answer: *"No, applying a raw Bollinger Band on MCX gold spreads is suicidal for two reasons: First, un-normalized spreads are distorted by differing purities and quotation units. Second, over 75% of 2-sigma events are false traps occurring in illiquid hours or failing Indian statutory fees. Our innovation is the deterministic purity transform combined with the Triple-Filter reality hurdle."*

Q3: "How do you handle Indian commodity taxes and slippage?"

Your Answer: *"We explicitly model the full Indian statutory fee schedule: Commodity Transaction Tax (CTT) at 0.01%, Stamp Duty at 0.002%, MCX Turnover fee at 0.0021%, GST at 18%, plus 3.5 basis points of bid-ask slippage per leg. If an anomaly's net expected reversion doesn't exceed our 14.0 bps hurdle, we discard it."*

Q4: "Is there any look-ahead bias in your backtest?"

Your Answer: *"Strictly zero. All rolling statistics (μ and σ) use trailing 20-day windows with no knowledge of future bars. Trades detected at bar close t are executed on bar open t+1. We also simulate automatic liquidation 5 days before contract expiry to prevent delivery notice liabilities."*